

1) School Management System

```
SELECT S.StudentName, C.ClassName, T.TeacherName, S.Grade FROM Student S
JOIN Class C ON S.ClassID = C.ClassID JOIN Teacher T ON C.TeacherID=T.TeacherID;
```

```
mysql>
mysql> INSERT INTO Performance VALUES
-> (1,1,4,3,'July'),
-> (2,2,4,3,'July'),
-> (3,3,4,3,'July'),
-> (4,4,4,1,'July'),
-> (5,5,3,4,'July');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT S.StudentName,
-> C.ClassName,
-> T.TeacherName,
-> S.Grade
-> FROM Student S
-> JOIN Class C
-> ON S.classID = C.classID
-> JOIN Teacher T
-> ON C.TeacherID = T.TeacherID;
+-----+-----+-----+-----+
| StudentName | ClassName | TeacherName | Grade |
+-----+-----+-----+-----+
| Anan        | X-A      | John        | A     |
| Priya       | X-A      | John        | B     |
| Rahul       | X-B      | Mary        | A     |
| Sneha       | X-C      | David       | A+    |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

2) Hospital Appointment System

```
SELECT D.DoctorName,
COUNT(A.AppointmentID) AS TotalAppointments
FROM Doctor D
JOIN Appointment A
ON D.DoctorID = A.DoctorID
WHERE MONTH(A.AppointmentDate) = MONTH(CURDATE())
AND YEAR(A.AppointmentDate) = YEAR(CURDATE())
GROUP BY D.DoctorID, D.DoctorName
HAVING COUNT(A.AppointmentID) > 10;
```

```

mysql> INSERT INTO Performance VALUES
-> (1,3,4,5,'July'),
-> (2,2,3,6,'July'),
-> (3,1,4,8,'July'),
-> (4,4,4,1,'July'),
-> (5,5,1,5,'July');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT S.StudentName,
-> C.ClassName,
-> T.TeacherName,
-> S.Grade
-> FROM Student S
-> JOIN Class C
-> ON S.ClassID = C.ClassID
-> JOIN Teacher T
-> ON C.TeacherID = T.TeacherID;
+-----+
| StudentName | ClassName | TeacherName | Grade |
+-----+
| Arun        | X-A      | John        | A     |
| Priya       | X-A      | John        | B     |
| Rahul       | X-B      | Mary        | A     |
| Sneha       | X-C      | David       | A+    |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT D.DoctorName,
-> COUNT(A.AppointmentID) AS TotalAppointments
-> FROM Doctor D
-> JOIN Appointment A
-> ON D.DoctorID = A.DoctorID
-> WHERE MONTH(A.AppointmentDate) = MONTH(CURRENT())
-> AND YEAR(A.AppointmentDate) = YEAR(CURRENT())
-> GROUP BY D.DoctorID, D.DoctorName
-> HAVING COUNT(A.AppointmentID) > 10;
Empty set (0.00 sec)

mysql>

```

3) E-Commerce Sub-query

```

SELECT ProductName
FROM Product
WHERE ProductID NOT IN
(
SELECT ProductID
FROM OrderDetails
);

```

```

MySQL 5.6 Command Line Client
-> FROM Student S
-> JOIN Class C
-> ON S.ClassID = C.ClassID
-> JOIN Teacher T
-> ON C.TeacherID = T.TeacherID;
+-----+
| StudentName | ClassName | TeacherName | Grade |
+-----+
| Arun        | X-A      | John        | A     |
| Priya       | X-A      | John        | B     |
| Rahul       | X-B      | Mary        | A     |
| Sneha       | X-C      | David       | A+    |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT D.DoctorName,
-> COUNT(A.AppointmentID) AS TotalAppointments
-> FROM Doctor D
-> JOIN Appointment A
-> ON D.DoctorID = A.DoctorID
-> WHERE MONTH(A.AppointmentDate) = MONTH(CURRENT())
-> AND YEAR(A.AppointmentDate) = YEAR(CURRENT())
-> GROUP BY D.DoctorID, D.DoctorName
-> HAVING COUNT(A.AppointmentID) > 10;
Empty set (0.00 sec)

mysql>
mysql> SELECT ProductName
-> FROM Product
-> WHERE ProductID NOT IN (
-> SELECT ProductID
-> FROM OrderDetails
-> );
+-----+
| ProductName |
+-----+
| Monitor     |
+-----+
1 row in set (0.00 sec)

mysql>

```

4) Relational Algebra

DBMSStudents $\leftarrow \pi_{\text{StudentID}} (\sigma_{\text{CourseName}='DBMS'} (\text{COURSE} \bowtie \text{ENROLLMENT}))$

OSStudents $\leftarrow \pi_{\text{StudentID}} (\sigma_{\text{CourseName}='OS'} (\text{COURSE} \bowtie \text{ENROLLMENT}))$

Result $\leftarrow \text{DBMSStudents} \cap \text{OSStudents}$

5) Payroll View

UPDATE Employee E
SET Salary = Salary * 1.10

WHERE Salary <

(

SELECT AvgSalary

FROM

(

SELECT DeptID, AVG(Salary) AS AvgSalary

FROM Employee

GROUP BY DeptID

) A

WHERE A.DeptID = E.DeptID

);

```
mysql> UPDATE Employee E
-> SET Salary = Salary * 1.10
-> WHERE Salary < (
-> SELECT AvgSalary
-> FROM
-> (
-> SELECT DeptID, AVG(Salary) AS AvgSalary
-> FROM Employee
-> GROUP BY DeptID
-> ) A
-> WHERE A.DeptID = E.DeptID
-> );

ERROR 1063 (HY000): You can't specify target table 'E' for update in FROM clause

mysql> CREATE TEMPORARY TABLE TempEmp AS
-> SELECT EmpID
-> FROM Employee E1
-> WHERE Salary < (
-> SELECT AVG(E2.Salary)
-> FROM Employee E2
-> WHERE E2.DeptID = E1.DeptID
-> );

Query OK, 2 rows affected (0.02 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> UPDATE Employee
-> SET Salary = Salary * 1.10
-> WHERE EmpID IN (
-> SELECT EmpID FROM TempEmp
-> );

Query OK, 2 rows affected (0.02 sec)
Rows matched: 2 Changed: 2 Warnings: 0

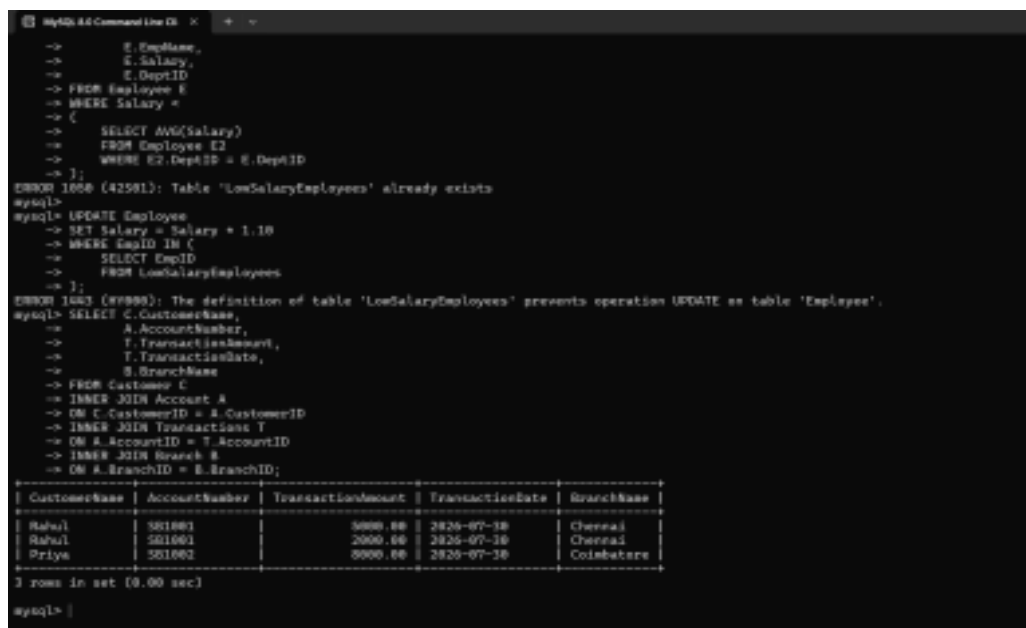
mysql> CREATE VIEW LowSalaryEmployees AS
-> SELECT E.EmpID,
-> E.Employee,
-> E.Salary,
-> E.DeptID
-> FROM Employee E
-> WHERE Salary < (
-> SELECT AVG(Salary)
-> FROM Employee E2
-> WHERE E2.DeptID = E.DeptID
-> );

ERROR 1050 (42501): Table 'LowSalaryEmployees' already exists

mysql>
mysql> UPDATE Employee
-> SET Salary = Salary * 1.10
-> WHERE EmpID IN (
```

6) Banking JOIN Operations

```
SELECT C.CustomerName,  
A.AccountNumber,  
T.TransactionAmount,  
B.BranchName  
FROM Customer C  
JOIN Account A  
ON C.CustomerID=A.CustomerID  
JOIN Transactions T  
ON A.AccountID=T.AccountID  
JOIN Branch B  
ON A.BranchID=B.BranchID;
```



```
--> E.EmpName,  
--> E.Salary,  
--> E.DeptID  
--> FROM Employee E  
--> WHERE Salary <  
--> (  
--> SELECT AVG(Salary)  
--> FROM Employee E2  
--> WHERE E2.DeptID = E.DeptID  
--> );  
ERROR 1050 (42581): Table 'LowSalaryEmployees' already exists  
mysql>  
mysql> UPDATE Employee  
--> SET Salary = Salary * 1.10  
--> WHERE EmpID IN (  
--> SELECT EmpID  
--> FROM LowSalaryEmployees  
--> );  
ERROR 1453 (HY000): The definition of table 'LowSalaryEmployees' prevents operation UPDATE on table 'Employee'.  
mysql> SELECT C.CustomerName,  
--> A.AccountNumber,  
--> T.TransactionAmount,  
--> T.TransactionDate,  
--> B.BranchName  
--> FROM Customer C  
--> INNER JOIN Account A  
--> ON C.CustomerID = A.CustomerID  
--> INNER JOIN Transactions T  
--> ON A.AccountID = T.AccountID  
--> INNER JOIN Branch B  
--> ON A.BranchID = B.BranchID;
```

CustomerName	AccountNumber	TransactionAmount	TransactionDate	BranchName
Rahul	SE1001	5000.00	2020-07-10	Chennai
Rahul	SE1001	2000.00	2020-07-10	Chennai
Priya	SE1002	8000.00	2020-07-10	Coimbatore

3 rows in set (0.00 sec)

```
mysql>
```

7) Retail Store GROUP BY

```
SELECT Category,  
SUM(SalesAmount) AS TotalSales  
FROM Sales  
GROUP BY Category  
HAVING SUM(SalesAmount) > 10000;
```

```

MySQL 5.6 Command Line Client
-> SET Salary = Salary * 1.10;
-> WHERE EmpID IN (
->   SELECT EmpID
->   FROM LowSalaryEmployees
-> );
ERROR 1063 (HY000): The definition of table 'LowSalaryEmployees' prevents operation UPDATE on table 'Employee'.
mysql> SELECT C.CustomerName,
->   A.AccountNumber,
->   T.TransactionAmount,
->   T.TransactionDate,
->   B.BranchName
-> FROM Customer C
-> INNER JOIN Account A
-> ON C.CustomerID = A.CustomerID
-> INNER JOIN Transactions T
-> ON A.AccountID = T.AccountID
-> INNER JOIN Branch B
-> ON A.BranchID = B.BranchID;
+-----+-----+-----+-----+-----+
| CustomerName | AccountNumber | TransactionAmount | TransactionDate | BranchName |
+-----+-----+-----+-----+-----+
| Rahul        | 551001        | 3000.00           | 2020-07-10      | Chennai    |
| Rahul        | 551001        | 2000.00           | 2020-07-10      | Chennai    |
| Priya        | 551002        | 5000.00           | 2020-07-10      | Coimbatore |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT Category,
->   SUM(SalesAmount) AS TotalSales
-> FROM Sales
-> GROUP BY Category
-> HAVING SUM(SalesAmount) > 10000;
+-----+-----+
| Category | TotalSales |
+-----+-----+
| Electronics | 13000.00 |
| Furniture   | 12000.00 |
+-----+-----+
2 rows in set (0.00 sec)

mysql>

```

8. Tuple Relational Calculus

$$\{m \mid \text{Member}(m) \wedge (\forall g)(\text{Genre}(g) \rightarrow \exists b(\text{Borrow}(b) \wedge b.\text{MemberID}=m.\text{MemberID} \wedge b.\text{Genre}=g.\text{GenreName}))\}$$

9) Inventory Constraints

```

CREATE TABLE Supplier(
  SupplierID INT PRIMARY KEY,
  SupplierName VARCHAR(50)
);

```

```

CREATE TABLE Inventory(
  ProductID INT PRIMARY KEY,
  ProductName VARCHAR(50),
  Quantity INT CHECK (Quantity >= 0),
  SupplierID INT,
  FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID) );

```

```

MySQL 4.0 Command Line 0.0.0.0
-> INNER JOIN Transactions T
-> ON A.AccountID = T.AccountID
-> INNER JOIN Branch B
-> ON A.BranchID = B.BranchID;
+-----+-----+-----+-----+-----+
| CustomerName | AccountNumber | TransactionAmount | TransactionDate | BranchName |
+-----+-----+-----+-----+-----+
| Rahul        | 501001        | 5000.00           | 2020-07-10     | Chennai    |
| Rahul        | 501001        | 2000.00           | 2020-07-10     | Chennai    |
| Priya        | 501002        | 8000.00           | 2020-07-10     | Coimbatore  |
+-----+-----+-----+-----+-----+
1 rows in set (0.00 sec)

mysql> SELECT Category,
-> SUM(SalesAmount) AS TotalSales
-> FROM Sales
-> GROUP BY Category
-> HAVING SUM(SalesAmount) > 10000;
+-----+-----+
| Category | TotalSales |
+-----+-----+
| Electronics | 13000.00 |
| Furniture   | 12000.00 |
+-----+-----+
2 rows in set (0.00 sec)

mysql> CREATE TABLE Supplier (
-> SupplierID INT PRIMARY KEY,
-> SupplierName VARCHAR(50)
-> );
ERROR 1050 (42501): Table 'supplier' already exists

mysql> CREATE TABLE Inventory (
-> ProductID INT PRIMARY KEY,
-> ProductName VARCHAR(50),
-> Quantity INT CHECK (Quantity >= 0),
-> SupplierID INT,
-> FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID)
-> );
ERROR 1050 (42501): Table 'inventory' already exists
mysql>

```

10. HR System

Create View for Monthly Attendance

```

CREATE VIEW MonthlyAttendance AS
SELECT EmpID,
COUNT(*) AS TotalPresent
FROM Attendance
WHERE Status = 'Present'
GROUP BY EmpID;

```

Create View for Performance Summary

```

CREATE VIEW PerformanceSummary AS
SELECT EmpID,
AVG(Rating) AS AverageRating
FROM Performance
GROUP BY EmpID;

```

Query the Views

```

SELECT * FROM MonthlyAttendance;

```

```

SELECT * FROM PerformanceSummary;

```

```
MySQL: 8.0 Command Line CS
-> };
ERROR 1050 (42501): Table 'Inventory' already exists
mysql> CREATE VIEW MonthlyAttendance AS
-> SELECT EmpID,
->        COUNT(*) AS TotalPresent
-> FROM Attendance
-> WHERE Status = 'Present'
-> GROUP BY EmpID;
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE VIEW PerformanceSummary AS
-> SELECT EmpID,
->        AVG(Rating) AS AverageRating
-> FROM Performance
-> GROUP BY EmpID;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM MonthlyAttendance;
+-----+-----+
| EmpID | TotalPresent |
+-----+-----+
| 1     | 2           |
| 2     | 1           |
| 4     | 1           |
+-----+-----+
3 rows in set (0.01 sec)

mysql>
mysql> SELECT * FROM PerformanceSummary;
+-----+-----+
| EmpID | AverageRating |
+-----+-----+
| 1     | 4.500000     |
| 2     | 3.000000     |
| 3     | 4.900000     |
| 4     | 4.100000     |
| 5     | 3.500000     |
+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```